

**PROJECT CONCEPT PROPOSAL***(To be accomplished by the researcher)*

Dear Proponent:

Thank you for applying to the DOST-GIA program. We value your proposal and your time for accomplishing the form. Approved projects are the government's investments on you and your institution. Your answers are very important to us as they may indicate your potential contributions to these public investments in research and development. We wish you all the best.

Sincerely,
DOST

I. CONCEPT PROJECT PROPOSAL SUMMARY INFORMATION

1. Project Title	Development of Functional Feed from Fish By-products for Livestock and Aquaculture Feeding
2. Project Duration <i>(in months)</i>	36 months
3. Estimated Budgetary Requirement	
4. Principal Research Question to be Addressed <i>(maximum of 100 words)</i>	

II. PROJECT LEADER PROFILE

1) Name	2) Position	3) Organization	4) Department/ Division	5) Contact Details <i>(telephone number, email)</i>
STEPHANIE L. PIMENTEL	ASSO. PROF 5	CAPIZ STATE UNIVERSITY	FOOD AND FISHERIES RESEARCH AND DEVELOPMENT CENTER	lrhc@capsu.edu.ph +639176399548
6) Other Ongoing Projects Being Handled by the Project Leader: _____ (number)				
Title of the Project		Funding Agency		Involvement in the Project

III. PROJECT PROFILE

1. Cooperating Agency/ies <i>(indicate role/responsibility of cooperating agency)</i>	FISHERIES BIOTECHNOLOGY CENTER LIVESTOCK BIOTECHNOLOGY CENTER		
2. Site (s) of Implementation <i>(Municipality/District/Province/ Region)</i>	Base Station: CAPIZ STATE UNIVERSITY		
	Other implementation sites:		
3. R&D Program	Basic		Applied

a)	Agriculture, Aquatic and Natural Resources		x
b)	Health		
c)	Industry, Energy and Emerging Technology		
d)	Disaster Risk Reduction and Climate Change Adaptation		

4. DOST Pillars Pursued <i>(Check all that apply)</i> ☐ Human Wellbeing ☐ Wealth Creation ☐ Wealth Protection ☒ Sustainability	(5) DOST Thematic Areas Covered <i>(Check all that apply)</i> ☐ Learning, Education, and Culture ☐ Health and Wellbeing ☒ Food and Agriculture ☐ Industrial Solutions and Competitiveness Development ☐ Transport and Mobility ☐ Energy and Utilities System ☐ Environment & Natural Resources ☐ Climate, Disaster Resilience, and Human Security ☐ Governance ☐ Others (please specify): _____
(6) Project Summary <i>(Please briefly describe your proposed research project ensuring the clarity of the scientific rationale; the deficit, needs, gaps or problems to be addressed; significance of the research and the solutions/benefits that it will offer; and relevance to the Philippine context. Include basis of problem identification (i.e., roadmap, value chain, industry request, etc.). Explicitly identify the objectives of the project and its target accomplishments and outputs. (maximum of 1,500 words).)</i>	
Fish processing activities typically generate 20–80 percent of solid waste, depending on the methodologies applied and the type of raw materials used. These solid wastes consist of the head, guts, viscera (including the liver and roe), bones, fins, scales, and skin. Fish by-products from fish markets represent a significant share of underutilized resources that may be used to produce protein-rich feed ingredients for aquamarine and livestock species (FAO, 2021). High-cost of conventional feed ingredients has been crippling the livestock, poultry and aquaculture industry. While fishmeal has been used as the common protein source feedstuff in feed formulation, it has been facing problems on availability, and escalating price per kg. Consequently, alternative methods of using fish by-products as a nutrient source and a medium for the development of functional feed has been considered. Fermentation has been a method associated with improving the nutritional value of feedstuffs for livestock and poultry feeding. Ensiling is a fermentation method that is commonly used for forages and fish waste. Several fish ensiling studies shown to improve the growth performance of livestock, poultry and aquaculture species. However, the majority of these, focused on the evaluation of the macronutrients of the fish silage. Only limited information has been published on the micronutrients, metabolites, and probiotic composition of the fish silage. These compounds and microorganisms can induce physiological benefits, enhance immunity and overall health of the animal and fish. Furthermore, utilizing fish silage for feed production can offer economic benefits by reducing the costs associated with waste disposal for the fishing industry. It creates an additional revenue stream by transforming waste into a valuable product. Converting this waste into functional feed for livestock and aquaculture can serve as a means to enrich the nutritional content of animal and aquaculture feeds. This study aligns with the principles of a circular economy by promoting the reuse and recycling of materials. It suggests a method to repurpose it into a beneficial product, closing the loop in the production cycle and reducing the environmental impact associated with waste disposal hence this study. Objectives: Generally this study aims to produce an alternative source of nutrients, extracellular products, and probiotics essential to sustain health and improve growth performance of livestock, poultry and aquaculture species. Specifically to: a. Evaluate the nutrient composition of the fish silage;	

- b. Identify and screen the potential probiotics present in the fish silage juice;
- c. Determine the extracellular products present in the silage juice; and
- d. Assess the effect of fish silage feeding and silage juice supplementation in the growth performance and immune response of poultry and tilapia and catfish feeding.

7)	SUBMITTED BY (Project Leader)
Signature	
Printed Name	
Designation/Title	
Date	

Note: If the project is part of a program, this form should be submitted together with the program concept proposal and one (1) curriculum vitae. Use A4 size paper, Arial typeface 10 points.

Proximate analysis:

A 100g sample will be dried and further analysed for its proximate analysis before and after ensiling following the method described in the AOAC 2022 standard in an outsourced laboratory.

Metagenomic Pipeline:

The microbial community of the fermented products will be profiled and characterized through metagenomics to clarify the product's organoleptic and microbial safety qualities, as well as the microbiome interactions of the fermented product. The Metagenomics analysis will be broken down into four main steps: (i) extraction of microbial DNA from the sample, (ii) library preparation, (iii) DNA sequencing, and (iv) bioinformatic processing and analysis of the generated sequence data.

Analysis of extracellular products in the fish silage juice